

Sanjith Shanmugavel

San Diego, CA | sshanmugavel@ucsd.edu | 408-431-2085 | [LinkedIn](#) | [Github](#)

EDUCATION

University of California San Diego

La Jolla, CA

B.S. in Math-Computer Science, Minor in Data Science, GPA: 3.58

Expected 2027

Relevant Coursework: Supervised/Unsupervised Machine Learning, Data Structures, Object Oriented Design

TECHNICAL EXPERTISE

Programming Languages: Python, TypeScript, JavaScript, Java, C, HTML, CSS, MATLAB, LaTeX
Frameworks & Libraries: React, FastAPI, Docker, Tailwind CSS, WebSockets, PyTorch, scikit-learn, D3.js, spaCy, YOLOv8, Pandas, NumPy, scipy, BrainFlow, Pygame
Tools & Technologies: AWS, NGINX, Git, PostgreSQL, GitHub, Docker Compose, REST-ful APIs, Linux, OpenAI API, Roboflow, Weights & Biases, Google Colab, Figma

EXPERIENCE

Product Engineering Intern

Aug. 2022 – Jan. 2023

Bumper Inc.

Palo Alto, CA

- Worked with engineers and designers to **design user interface** for “Bumper Bites”, an **in-app investing education system** reaching **16,000+ users**, delivering bite-sized informational blogs and videos
- Developed **content management logic** enabling flexible updates to educational material without redeploying app
- Collaborated with product managers, designers, and backend engineers under **Agile methodology** to create **modular components and UI elements** for blog and video sections

Technical Director

Sept. 2024 – Present

Triton Neurotech

San Diego, CA

- Direct a team of **70+ developers and researchers** in building brain-computer interfaces for assistive technologies, including **prosthetic hand, RC car, and autonomous wheelchair control systems**
- Built scalable **ML pipelines** across custom hardware using **sklearn, scipy, pandas, NumPy, and BrainFlow**
- Raised **\$3,000+ in sponsorships**, instructed **120+ students** through ML and BCI bootcamps, and earned **2nd Place at the 2025 CA Neurotech Conference** for a **wearable biosignal suit and companion app**

PROJECTS

[AgentFBA](#) – Agentic Market Research Platform (Full-Stack, Deployed)

June 2025 – Present

React, TypeScript, FastAPI, PostgreSQL, Python, D3.js, Tailwind CSS, Docker, NGINX, AWS EC2, Ubuntu

- Built a suite of **agentic market research tools** for Amazon sellers to analyze **500M+ products** through **natural language queries**, algorithmic **opportunity scoring**, and **cluster visualization**
- Engineered scalable **FastAPI backend** with **database integration** and **<8s response times** via LLM-driven query to product list pipeline that verifies responses with **Pydantic models** and **schema enforcement**
- Developed responsive frontend using **React + TypeScript** with **D3.js circle mapping** and **Tailwind CSS** UI, deployed on **AWS EC2** via **Docker Compose** and **NGINX**

[Iatric](#) – AI Medical Assistant (Full-Stack)

July 2025 – Sept. 2025

FastAPI, WebSockets, Python, Groq Whisper, spaCy, LLMs, TypeScript, Tailwind CSS

- Built **real-time AI medical companion** that transcribes doctor-patient conversations and evaluates medical claims against **trusted research papers** to help prevent misdiagnosis
- Developed backend with **WebSockets, FastAPI, and Python** for live audio streaming via **Groq Whisper**, implemented custom **NLP pipeline** with **spaCy** and **link phrasing** for claim extraction
- Designed frontend with **TypeScript + Tailwind CSS** and integrated **Google Gemini AI SDK** and **PubMed** to verify claims, using **Presidio Analyzer/Anonymizer** to ensure patient confidentiality

[Recycling Sentry](#) – Autonomous Object Recognition Trash Robot

Nov 2023 – Jul 2024

PyTorch, YOLOv8, Roboflow, Python, Google Colab, Weights & Biases

- Developed **image-recognition AI** to reduce recycling contamination by categorizing trash by disposal method
- Annotated **4,000+ data points** using **Roboflow**, and **trained YOLOv8 model in Pytorch** to achieve **85.2% precision** across 6 material classes
- Recognized at multiple competitions, including Amgen State Hackathon (**4th Place, Best Presentation**), Conrad-Equinor Challenge (**Semi-Finalist**), and Blue Ocean Pitch Competition (**Top 1% worldwide**)